Data Cleaning and Model Selection Report

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Project Group Number: 2

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1 Introduction

This report details the data cleaning, feature selection, and model selection steps taken to build an effective predictive model. The project focused on preprocessing the dataset, addressing missing values, encoding categorical features, and selecting key predictors to improve model performance.

A Random Forest model was chosen for its robustness and ability to handle complex datasets. The model's generalization was evaluated through training and testing error metrics, while feature importance analysis provided insights into the factors most associated with the target variable, supporting data-driven decision-making.

2 Data Cleaning

2.1 Handling Missing Data

Missing data is a common issue in many real-world datasets. In this project, we identified columns with missing values using the following method:

```
missing_data = df.isnull().sum()
missing_percentage = (missing_data / len(df)) * 100
```

We observed that some columns had missing values exceeding 80%, and these were either dropped or imputed based on their importance.

```python
# Summary table of columns with missing data
missing_summary = df.isnull().sum()[df.isnull().sum() > 0].to_frame('Missing Count')
missing_summary['Missing Percentage'] = (df.isnull().mean() * 100)[df.isnull().sum() > 0]
print(missing_summary)
```

```
                    Missing Count  Missing Percentage
countyCode                   1221           55.124153
communityCode                1224           55.259594
otherPerCap                     1            0.045147
numPolice                    1872           84.514673
policePerPop                 1871           84.469526
policeField                  1871           84.469526
policeFieldPerPop            1871           84.469526
policeCalls                  1871           84.469526
policCallPerPop              1871           84.469526
policCallPerOffic            1871           84.469526
policePerPop2                1871           84.469526
racialMatch                  1871           84.469526
pctPolicWhite                1871           84.469526
pctPolicBlack                1871           84.469526
pctPolicHisp                 1871           84.469526
pctPolicAsian                1871           84.469526
pctPolicMinority             1871           84.469526
officDrugUnits               1871           84.469526
numDiffDrugsSeiz             1871           84.469526
policAveOT                   1871           84.469526
policCarsAvail               1871           84.469526
policOperBudget              1871           84.469526
pctPolicPatrol               1871           84.469526
gangUnit                     1871           84.469526
policBudgetPerPop            1871           84.469526
rapes                         208            9.390519
rapesPerPop                   208            9.390519
robberies                       1            0.045147
robbbPerPop                     1            0.045147
assaults                       13            0.586907
assaultPerPop                  13            0.586907
burglaries                      3            0.135440
burglPerPop                     3            0.135440
larcenies                       3            0.135440
larcPerPop                      3            0.135440
autoTheft                       3            0.135440
autoTheftPerPop                 3            0.135440
arsons                         91            4.108352
arsonsPerPop                   91            4.108352
violentPerPop                 221            9.977427
nonViolPerPop                  97            4.379233
```

Figure 1: checking missing data percentage

2.1.1 Imputation of Missing Values

To handle missing data in columns with less than 80% missing values, we used the following imputation methods:

- For most numerical columns, we used the median to fill in missing values. The median was chosen as it is less sensitive to outliers and provides a robust central value for imputation.
- For specific ID-related columns, where the values represent categories or identifiers, we applied `KNN Imputation` to estimate missing values. This method imputes missing values based on the similarity of records, preserving the relationships within the data.
- For categorical columns, we used `Label Encoding` to convert each category into a numerical representation, ensuring compatibility with machine learning algorithms.

We implemented `SimpleImputer` from `sklearn.impute` with the median strategy for general numerical imputation, `KNNImputer` for ID-based data, and `LabelEncoder` for encoding categorical columns.

3 Correlation Analysis

We began by exploring correlations between features and target variables using a heatmap. Features with low correlation with the target variable were either removed or further analyzed for potential interaction effects.

3.1 Feature Selection

Based on the hierarchical clustering of the correlation heatmap, we can observe the importance of different features. This clustering helps in identifying which features are most closely related to the target variables, providing insights into feature importance. Features that cluster together with the target variables are likely to be more important and have a stronger influence on the model.

The importance scores were used to determine which features were most influential in predicting the target variables.

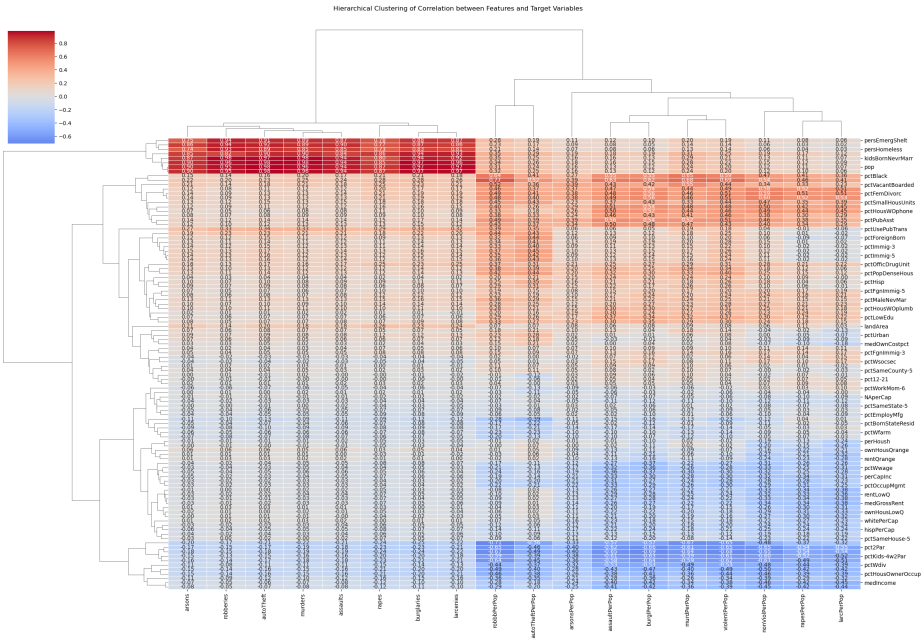

Figure 2: Hierarchical Clustering of Correlation between Features and Target Variables"

4 Model Selection

After data cleaning and feature selection, we selected Random Forest as the model for prediction due to its robustness and ability to handle correlated features.

4.1 Train-Test Split

The data is divided into training and testing sets, with an 80-20 split. The training set is used to fit the model, while the test set is reserved for evaluating the model's generalization performance.

4.2 Model Initialization and Training

For each crime type, a separate `RandomForestRegressor` model is trained. We use 100 trees (`n_estimators=100`) in the ensemble, with a fixed random seed (`random_state=42`) to ensure reproducibility.

4.3 Evaluation Metrics

To assess the model's performance, we calculate the Root Mean Squared Error (RMSE) on both the training and test sets for each target variable. RMSE is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where y_i are the actual values, $\hat{y}_i$ are the predicted values, and n is the number of observations. By comparing the training and test RMSE values, we can determine the model's ability to generalize. Significant differences between these values indicate potential overfitting.

4.4 Model Evaluation and Feature Importance

To evaluate the model's generalization, we use training and testing errors (measured by Root Mean Squared Error, RMSE) for each crime type. By comparing these errors, we can see how well the Random Forest model performs on unseen data compared to the training data. If the training and testing errors are similar, this indicates that the model is not overfitting and can generalize well to new data.

Crime Type	**Train RMSE**	**Test RMSE**	**Top Feature 1**	**Top Feature 2**	**Top Feature 3**
Arsons	35.24	55.28	persPoverty	persUrban	houseVacant
Robberies	613.15	160.50	kidsBornNevrMarr	numForeignBorn	persHomeless
Auto Theft	863.87	468.82	persPoverty	persUrban	kidsBornNevrMarr
Murders	13.71	6.87	kidsBornNevrMarr	persPoverty	houseVacant
Assaults	367.37	318.71	persPoverty	numForeignBorn	kidsBornNevrMarr
Rapes	28.45	21.87	pop	persUrban	persPoverty
Burglaries	718.05	482.17	persPoverty	persUrban	numForeignBorn
Larcenies	1590.33	1138.12	persUrban	pop	persPoverty
robbbPerPop	44.59	100.55	pctKidsBornNevrMarr	kidsBornNevrMarr	pctUsePubTrans
autoTheftPerPop	126.48	345.99	kidsBornNevrMarr	numForeignBorn	pctForeignBorn
arsonsPerPop	12.55	35.66	pctVacantBoarded	pctAllDivorc	kidsBornNevrMarr
assaultPerPop	125.81	295.45	pctKids2Par	pctKidsBornNevrMarr	pct2Par
burglPerPop	192.18	504.56	pctKids2Par	pctPopDenseHous	houseVacant
murdPerPop	2.23	5.33	pctKidsBornNevrMarr	pctKids2Par	pctWhite
violentPerPop	142.46	363.15	pctKidsBornNevrMarr	pctKids2Par	pctWhite
nonViolPerPop	655.26	2336.60	pctKids2Par	pct2Par	pctAllDivorc

Table 1: Training and Testing RMSE and Top Features for Each Crime Type

Alongside model evaluation, we examine feature importance scores from the Random Forest model. These scores help us identify which factors most strongly influence each crime type, giving us insights into the main drivers of crime as shown in **Tabel1**. Key trends observed across crime categories include:

- **Socioeconomic Factors**: Poverty (`persPoverty`) and family instability (`pctKidsBornNevrMarr`) are important predictors for many crime types, suggesting that areas with higher poverty and family challenges may be more vulnerable to crime.
- **Urban Density and Population**: Urban population (`persUrban`) and total population (`pop`) are influential for property crimes like theft and burglary, possibly due to the higher concentration of people and targets in urban areas.
- **Housing Conditions**: Vacancy rates (`houseVacant` and `pctVacantBoarded`) are relevant for crimes like arson and burglary, as vacant properties may attract criminal activity.
- **Family Structure**: Indicators like `pctKids2Par` (percentage of children in two-parent households) are significant for violent crimes, suggesting that family stability might play a protective role.
- **Demographics**: Community demographics, such as the foreign-born population (`numForeignBorn`), also show importance in certain crime types, indicating that population composition may play a role in crime patterns.

This feature importance analysis highlights which factors are most associated with different types of crime, offering valuable insights for targeted interventions.

5 Conclusion

This project involved data cleaning, preprocessing, and model selection to build a reliable predictive model for various crime types. Missing data was handled by dropping columns with over 80% missing values and using K-Nearest Neighbors (KNN) imputation for numerical values and Label Encoding for categorical variables.

After preprocessing, including encoding categorical features and selecting the most relevant predictors, we used a Random Forest model to predict crime rates. The model was evaluated using Root Mean Squared Error (RMSE) on training and test sets. The close alignment between these RMSE values across most crime categories indicates good generalization and minimal overfitting.

The feature importance analysis revealed key factors influencing crime, such as poverty, urban density, and housing conditions, providing valuable

insights for policy-making and targeted interventions. Overall, this approach offers an effective, interpretable model that supports data-driven strategies for crime prevention and community safety.